

Inverse Reinforcement Learning for Mean-field Games with Average Reward Criterion

Şevket Kaan Alkır and Naci Saldi

Abstract—We study the inverse reinforcement learning (IRL) problem for discrete-time, infinite-horizon mean-field games (MFGs) with an average-reward criterion. Unlike the forward setting, where the reward function is known, IRL assumes access only to expert demonstrations that are optimal under some unknown reward function. The objective is to recover a policy that explains the expert behavior under some unknown reward, selected via the maximum causal entropy principle. Our approach is based on the maximum causal entropy principle, which selects the least biased policy among those consistent with the observed demonstrations. We show that the resulting non-convex formulation is equivalently reformulated as a convex optimization problem over occupation measures. Furthermore, we establish that the dual objective is smooth and strongly convex over compact sets and derive a variational representation using a log-partition formulation. Finally, we propose a first-order algorithm for solving the dual problem and recovering an entropy-maximizing equilibrium policy.

Index Terms—Inverse reinforcement learning, mean-field games, maximum entropy principle, average-reward criterion.

I. INTRODUCTION

Mean-field games (MFGs) were introduced to study strategic interactions in large populations, where each individual's decision depends on the overall behavior of the group rather than direct interactions with every other agent [1, 2]. In stationary settings, agents consider a fixed distribution of states [3], effectively reducing the problem to a Markov decision process (MDP) constrained by this stationary distribution. The equilibrium concept, known as the stationary mean-field equilibrium (MFE), consists of a policy and a state distribution that satisfy the Nash Certainty Equivalence principle [1]. Under reasonable assumptions, the existence of such an equilibrium follows from Kakutani's fixed-point theorem, and as the number of agents grows, the resulting policy closely approximates a Nash equilibrium [4].

When the reward function is known, traditional MFG methods can be used to compute equilibria, and recent studies have incorporated reinforcement learning techniques into this

framework [5]. However, defining an appropriate reward function is often challenging in practice. Inverse reinforcement learning (IRL) tackles this issue by learning the reward function from expert demonstrations. By uncovering the underlying objectives driving expert behavior, IRL improves interpretability and can enhance the model's ability to generalize to new environments [6].

Recent studies have explored IRL in the context of MFGs. [7] reformulate a specific MFG as an MDP and apply maximum entropy IRL, but this approach applies only to fully cooperative settings where all agents share a common reward. In contrast, [8] extend IRL to decentralized, non-cooperative MFGs using a maximum margin method. [9] further refine this by introducing mean-field adversarial IRL, combining elements of decentralized IRL, maximum entropy IRL, and generative adversarial learning.

Imitation learning offers a complementary perspective by directly learning policies from demonstrations, bypassing reward recovery. IL has also been applied in the MFG context [10], especially when reward functions are difficult to define.

Previous studies have focused on finite-horizon costs, leading to convex optimization formulations based on classical maximum entropy and maximum margin methods. However, the classical maximum entropy approach in [9] breaks down in infinite-horizon settings, as the state-action distribution over the path space becomes ill-defined. To address this, [11] introduce the maximum causal entropy principle for infinite-horizon MDPs using state-action occupation measures. Building on this, we extend its application to infinite-horizon MFGs under the average-reward criterion. In a related work, [12], where the second author of this paper applied the same occupation measure-based formulation, successfully solved the IRL problem for MFGs under a discounted reward framework. Here, we further develop this approach for the average-reward setting.

II. MEAN-FIELD GAMES: FORWARD REINFORCEMENT LEARNING

A discrete-time stationary MFG is defined by the tuple $(\mathcal{X}, \mathcal{A}, p, r)$, where $\mathcal{X}$ and $\mathcal{A}$ denote the finite state and action spaces, respectively. The transition probability function $p : \mathcal{X} \times \mathcal{A} \times \mathcal{P}(\mathcal{X}) \rightarrow \mathcal{P}(\mathcal{X})$ determines how the next state evolves, while the one-stage reward function $r : \mathcal{X} \times \mathcal{A} \times \mathcal{P}(\mathcal{X}) \rightarrow [0, \infty)$ specifies the immediate reward received. Given a current state x_t , action a_t , and state-measure μ , the reward is given by $r(x_t, a_t, \mu)$, and the next state is

Şevket Kaan Alkır and Naci Saldi are with Department of Mathematics at Bilkent University, Ankara, Turkey, Emails: kaan.alkir@bilkent.edu.tr, naci.saldi@bilkent.edu.tr. This work was supported by the Scientific and Technological Research Council of Turkey (TUBITAK), under Grant no: 1001-124F134.

© 2025 IEEE. Personal use of this material is permitted. Permission from IEEE must be obtained for all other uses, in any current or future media, including reprinting/republishing this material for advertising or promotional purposes, creating new collective works, for resale or redistribution to servers or lists, or reuse of any copyrighted component of this work in other works.

drawn from the probability distribution: $x_{t+1} \sim p(\cdot | x_t, a_t, \mu)$. In an MFG, the state-measure $\mu \in \mathcal{P}(\mathcal{X})$ represents the collective behavior of all other agents. Intuitively, μ captures the limiting distribution of the empirical state distribution as the number of agents approaches infinity. While classical MFG models typically consider a time-dependent sequence of state-measures $\{\mu(t)\}$, our focus is on the stationary setting, where the population distribution remains constant over time, i.e., $\mu(t) = \mu$ for all t .

To fully describe the model dynamics, we must also specify how the agent selects actions. A *policy* π is a conditional distribution over $\mathcal{A}$ given $\mathcal{X}$, meaning that $\pi : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{A})$. Let Π denote the set of all such policies.

Before introducing the optimality concept in MFGs, we first define the average reward associated with a given policy under a fixed state measure. In average MFGs, for a fixed μ , the average reward of a policy π is given by

$$V_\mu(\pi, \mu_0) = \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}^\pi [r(x_t, a_t, \mu)],$$

where the initial state x_0 follows the distribution μ_0 .

To formalize the notion of optimality, we define the set-valued mapping $\Phi : \mathcal{P}(\mathcal{X}) \rightarrow 2^\Pi$, where 2^Π denotes the collection of all subsets of Π , as follows: $\Phi(\mu) = \{\hat{\pi} \in \Pi : V_\mu(\hat{\pi}, \mu) = \sup_{\pi \in \Pi} V_\mu(\pi, \mu)\}$. Namely, $\Phi(\mu)$ represents the set of optimal policies for a given state measure μ .

Similarly, we define the set-valued mapping $\Lambda : \Pi \rightarrow 2^{\mathcal{P}(\mathcal{X})}$, which describes how a policy determines a stationary state distribution. Specifically, for any $\pi \in \Pi$, the state-measure $\mu_\pi \in \Lambda(\pi)$ is an invariant distribution of the transition dynamics, satisfying

$$\mu_\pi(\cdot) = \sum_{x \in \mathcal{X}} p(\cdot | x, a, \mu_\pi) \pi(a | x) \mu_\pi(x).$$

We define the equilibrium concept for the MFG as follows:

Definition 1. A pair $(\pi^*, \mu^*) \in \Pi \times \mathcal{P}(\mathcal{X})$ is called an MFE if $\pi^* \in \Phi(\mu^*)$ and $\mu^* \in \Lambda(\pi^*)$.

In the *forward reinforcement learning (RL) problem* for MFGs, the objective is to compute an MFE when the model is known or to learn the equilibrium when the model is unknown. This involves designing algorithms that either solve for the equilibrium policy directly using the given transition dynamics and reward structure or infer the equilibrium behavior from observed interactions.

In contrast, the *IRL problem* for MFGs takes an alternative perspective. Here, the goal is to *recover a policy that is consistent with expert demonstrations*, which are assumed to be optimal for some unknown reward structure. Instead of computing the optimal policy from a given reward, the task is to infer a policy that best explains the observed expert behaviors.

The inverse setting plays a crucial role in applications where the reward function is not explicitly known but must be deduced from observed decision-making patterns, such as in economics, autonomous systems, and multi-agent learning. In the next section, we formally introduce this setting and

develop a framework to recover a policy consistent with expert demonstrations.

III. INVERSE REINFORCEMENT LEARNING

Since IRL aims to infer the reward function from expert demonstrations, it is essential to restrict the space of possible rewards to make the problem more tractable. A widely used assumption in IRL for MFGs is that the reward function can be expressed as a linear combination of feature vectors, which depend on the state, action, and mean-field term:

$$\mathcal{R} := \{r(x, a, \mu) = \langle \theta, f(x, a, \mu) \rangle \mid \theta \in \mathbb{R}^k, \\ f : \mathcal{X} \times \mathcal{A} \times \mathcal{P}(\mathcal{X}) \rightarrow \mathbb{R}^k\}$$

where $f(x, a, \mu) \in \mathbb{R}^k$ is the feature vector that includes information about the state-action pair (x, a) and the mean-field term μ .

Assumption 1. We assume that for any policy π and mean-field term μ , the state process is positive Harris recurrent and aperiodic, guaranteeing the convergence of empirical averages [13, Thm. 13.3.3]. In other words, the state process exhibits ergodic behavior under any policy and mean-field term, ensuring long-run statistical stability.

Under these assumptions, we can replace the $\limsup$ operation with $\lim$ throughout the remainder of the paper, as ergodicity ensures the existence of time-average limits.

In IRL environment, we assume that under some MFE (π_E, μ_E) , some experts generate the following trajectories: $\mathcal{Z} = \left\{ (x_t^i, a_t^i)_{t \geq 0} \right\}_{i=1}^d$. Since μ_E is invariant probability distribution under policy π_E , when the mean-field term μ_E is in state dynamics, under Assumption 1, ergodic theorem implies that

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \left(\frac{1}{d} \sum_{i=1}^d \mathbf{1}_{\{x_t^i = x\}} \right) = \mu_E(x), \quad \forall x \in \mathcal{X}.$$

While it is sufficient to consider only one path in $\mathcal{Z}$, obtaining more concrete estimate of μ_E is possible by considering all paths. Also, if d is large enough, we can obtain an estimate for the feature expectation vector as follows:

$$\frac{1}{d} \sum_{i=1}^d \left(\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} f(x_t^i, a_t^i, \mu_E) \right) \simeq \langle f \rangle_{\pi_E, \mu_E}$$

where the average-reward expected feature vector under the MFE (π_E, μ_E) for $x_0 \sim \mu_E$ is defined as,

$$\langle f \rangle_{\pi_E, \mu_E} := \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}^{\pi_E, \mu_E} [f(x_t, a_t, \mu_E)]. \quad (1)$$

Thus, for the remainder of this paper, we assume that the average feature expectation vector $\langle f \rangle_{\pi_E, \mu_E}$ under the expert MFE (π_E, μ_E) and the mean-field term μ_E are given.

Remark 1. In the IRL literature for Markov decision processes (MDPs), it is common to assume that the feature expectation vector is known. Extending this idea to MFGs, recent studies ([8, 9]) show that an unbiased estimate of the mean-field term can be obtained from expert trajectories. By

the ergodic theorem, this estimate converges to the true mean-field term with probability 1. Hence, assuming that the exact mean-field term μ_E is known does not introduce a significant practical limitation. Using estimates instead of true quantities results in an approximate MFE but does not fundamentally alter the problem structure, aside from introducing estimation errors. Since these errors follow a straightforward analysis and offer little additional insight, we do not examine them further.

IV. MAXIMUM CAUSAL ENTROPY PRINCIPLE

In this section, we formulate the IRL problem as an optimization task that maximizes the causal entropy of the policy while satisfying certain constraints. These constraints ensure that the resulting policy not only constitutes an MFE but also explains the observed expert behavior.

Our primary objective is to identify a policy that best explains the expert's decisions while maximizing entropy. This approach favors the least biased policy among those that align with the expert demonstrations. To formalize this idea, we first define the average-reward causal entropy of a given policy.

Definition 2. Average-reward causal entropy $H(\pi)$ of the policy $\pi \in \Pi$ is defined as follows

$$H(\pi) := \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}^{\pi, \mu_E} [-\log \pi(a_t | x_t)].$$

In view of this definition, we can now define the average-reward maximum causal entropy IRL problem $(\text{OPT})_1$ as follows:

$$\begin{aligned} & \max_{\pi \in \mathbb{R}^{\mathcal{A} \times \mathcal{X}}} H(\pi) \\ \text{s.t. } & \pi(a | x) \geq 0, \quad \forall (x, a) \in \mathcal{X} \times \mathcal{A} \\ & \sum_{a \in \mathcal{A}} \pi(a | x) = 1, \quad \forall x \in \mathcal{X} \\ & \mu_E(x) = \sum_{(a, y)} p(x | y, a, \mu_E) \pi(a | y) \mu_E(y), \quad \forall x \in \mathcal{X} \\ & \langle f \rangle_{\pi, \mu_E} = \langle f \rangle_{\pi_E, \mu_E} \end{aligned}$$

In the optimization problem above, the first two constraints ensure that π belongs to Π . The third and fourth constraints guarantee that the pair (π, μ_E) forms an MFE, as explained in the following discussion.

In the MFG problem, the expert's behavior is governed by the MFE (π_E, μ_E) under some unknown reward function $r_E(x, a, \mu) = \langle \theta_E, f(x, a, \mu) \rangle$. This implies that π_E is the optimal policy for μ_E under the reward function r_E , and μ_E is the stationary state distribution under the policy π_E . Since there are typically multiple possible values of θ_E that could explain the expert's behavior, we address this ambiguity by applying the maximum causal entropy principle. Introduced in [14], this principle suggests that when multiple explanations are possible, the policy with the highest causal entropy should be preferred. This approach minimizes bias, except for the bias introduced by the feature expectation constraint.

Let π^* be a solution of $(\text{OPT})_1$. From the third constraint, μ_E is the stationary distribution under π^* , i.e., $\mu_E \in \Lambda(\pi^*)$. The fourth constraint gives $\langle f \rangle_{\pi, \mu_E} = \langle f \rangle_{\pi_E, \mu_E}$. But note that for any reward function $r_\theta(x, a, \mu) =$

$\langle \theta, f(x, a, \mu) \rangle$, we can write the average reward of any policy under the mean-field term μ as follows: $V_\mu(\pi, \mu) = \langle \theta, \langle f \rangle_{\pi, \mu_E} \rangle$. Since π_E is optimal for μ_E with respect to $r_E(x, a, \mu) = \langle \theta_E, f(x, a, \mu) \rangle$, from the forth constraint, we have $V_{\mu_E}(\pi^*, \mu_E) = V_{\mu_E}(\pi_E, \mu_E) = \sup_{\pi \in \Pi} V_{\mu_E}(\pi, \mu_E)$. It follows that $\pi^* \in \Phi(\mu_E)$. Thus, (π^*, μ_E) forms an MFE, and solving $(\text{OPT})_1$ leads to an MFE.

V. CONVEX REFORMULATION OF $(\text{OPT})_1$

Note that $(\text{OPT})_1$ is not a convex optimization problem, as the final constraint (i.e., matching average-reward feature expectations) is not convex in π . To reformulate the problem as a convex optimization problem, we employ state-action occupation measures induced by policies. This technique, used in [11] to address the IRL problem in infinite-horizon discounted MDPs, helps achieve convexity.

Definition 3. We define the state-action occupation measure ν_π for any policy $\pi \in \Pi$ as

$$\nu_\pi(x, a) := \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}^{\pi, \mu_E} [\mathbf{1}_{\{(x_t, a_t) = (x, a)\}}],$$

and corresponding state occupation measure as

$$\nu_\pi^{\mathcal{X}}(x) := \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}^{\pi, \mu_E} [\mathbf{1}_{\{x_t = x\}}] = \sum_{a \in \mathcal{A}} \nu_\pi(x, a).$$

Notice that ν_π is a probability measure for any policy $\pi \in \Pi$.

To derive the convex reformulation of $(\text{OPT})_1$, we will begin by stating several lemmas below.

Lemma 1. Suppose $\pi \in \Pi$ is a feasible point for the optimization problem $(\text{OPT})_1$. Then, for all $x \in \mathcal{X}$, we have $\nu_\pi^{\mathcal{X}}(x) = \mu_E(x)$, and furthermore,

$$\nu_\pi^{\mathcal{X}}(x) = \sum_{(y, a) \in \mathcal{X} \times \mathcal{A}} p(x | y, a, \mu_E) \nu_\pi(y, a).$$

Lemma 2. If $\pi \in \Pi$ is a feasible point for $(\text{OPT})_1$, then $H(\pi)$ can be written as

$$H(\pi) = \sum_{(x, a) \in \mathcal{X} \times \mathcal{A}} -\log \left(\frac{\nu_\pi(x, a)}{\mu_E(x)} \right) \nu_\pi(x, a).$$

Lemma 3. The average-reward feature expectation vector can be written as

$$\langle f \rangle_{\pi, \mu_E} = \sum_{(x, a) \in \mathcal{X} \times \mathcal{A}} f(x, a, \mu_E) \nu_\pi(x, a).$$

The proofs follow directly from stationarity, ergodicity, and the Definition 3, and are thus left out for brevity.

In view of above results, let us now reformulate $(\text{OPT})_1$ in terms of state-action occupation measure. This new formulation is denoted by $(\text{OPT})_2$, which will be shown to be convex and yield the same solution as $(\text{OPT})_1$:

$$\begin{aligned} & \max_{\nu \in \mathbb{R}^{\mathcal{X} \times \mathcal{A}}} \sum_{(x, a) \in \mathcal{X} \times \mathcal{A}} -\log \left(\frac{\nu(x, a)}{\mu_E(x)} \right) \nu(x, a) \\ \text{s.t. } & \sum_{(x, a) \in \mathcal{X} \times \mathcal{A}} f(x, a, \mu_E) \nu(x, a) = \langle f \rangle_{\pi_E, \mu_E} \\ & \mu_E(x) = \sum_{(y, a) \in \mathcal{X} \times \mathcal{A}} p(x | y, a, \mu_E) \nu(y, a), \quad \forall x \in \mathcal{X} \\ & \nu^{\mathcal{X}}(x) = \mu_E(x), \quad \forall x \in \mathcal{X} \\ & \nu(x, a) \geq 0, \quad \forall (x, a) \in \mathcal{X} \times \mathcal{A}, \end{aligned}$$

Here, we reformulate the problem with respect to the set of occupation measures instead of policies in II. The following result establishes the convexity of $(\text{OPT})_2$.

Lemma 4. $(\text{OPT})_2$ is a convex optimization problem.

Proof. We can rewrite the objective function as

$$\sum_{(x,a) \in \mathcal{X} \times \mathcal{A}} -\log(\nu(x,a)) \nu(x,a) + \sum_{(x,a) \in \mathcal{X} \times \mathcal{A}} \log(\mu_E(x)) \nu(x,a).$$

The first term is the entropy of ν , which is strongly concave in ν [15, p. 37]; the second term is linear in ν . Hence the full objective is strongly concave. All constraints in $(\text{OPT})_2$ are linear in ν . $\square$

Remark 2. By the duality theory, the optimization problem $(\text{OPT})_2$ can be rewritten in a min-max form. Indeed, introducing the penalty terms $\alpha \in \mathbb{R}^k$ and $\beta, \theta \in \mathbb{R}^{\mathcal{X}}$ to enforce constraints, we can express $(\text{OPT})_2$ as:

$$\max_{\nu \in \mathcal{P}(\mathcal{X} \times \mathcal{A})} \min_{\substack{\alpha \in \mathbb{R}^k \\ \beta, \theta \in \mathbb{R}^{\mathcal{X}}}} \left\{ H(\nu) + \sum_{(x,a) \in \mathcal{X} \times \mathcal{A}} k_{\alpha, \beta, \theta}(x,a) \nu(x,a) - \langle \alpha, \langle f \rangle_{\pi_E, \mu_E} \rangle - \sum_{x \in \mathcal{X}} \theta_x \mu_E(x) \right\},$$

where $k_{\alpha, \beta, \theta}(x,a) := \log(\mu_E(x)) + \langle \alpha, f(x,a, \mu_E) \rangle + \theta_x + \sum_{z \in \mathcal{X}} \beta_z (p(z | x,a, \mu_E) - \mu_E(z))$. Here, $H(\nu)$, as the entropy of ν , is strongly concave in ν , while the second term is linear in ν . The function $k_{\alpha, \beta, \theta}(x,a)$ and the remaining terms are linear in α, β, θ , guarantees the convexity of the objective in α, β, θ . This min-max formulation will be used to obtain the dual of the optimization problem $(\text{OPT})_2$ in the next section.

Theorem 1. There are mappings between feasible points of $(\text{OPT})_1$ and $(\text{OPT})_2$, and under these mappings two equivalent feasible points lead to the same objective value.

Sketch of Proof. ($\Rightarrow$) Let $\pi \in \Pi$ be feasible for $(\text{OPT})_1$, and let $\nu_\pi(x,a) := \mu_E(x) \pi(a | x)$. By Lemma 1, this measure satisfies the stationarity condition of $(\text{OPT})_2$, and by Lemma 2, satisfy the feature matching constraint. Since ν_π is a probability distribution and all constraints are met, ν_π is feasible for $(\text{OPT})_2$. The equality of objective functions follows from Lemma 2.

($\Leftarrow$) Let $\nu \in \mathcal{P}(\mathcal{X} \times \mathcal{A})$ be feasible for $(\text{OPT})_2$, with $\nu_{\mathcal{X}} = \mu_E$. Define the policy $\pi_\nu(a | x) := \frac{\nu(x,a)}{\mu_E(x)}$. Using this definition, the stationarity constraint becomes

$$\mu_E(x) = \sum_{y,a} p(x | y,a, \mu_E) \pi_\nu(a | y) \mu_E(y),$$

so $\mu_E \in \Lambda(\pi_\nu)$. Moreover, the occupation measure $\nu_{\pi_\nu}(x,a) = \mu_E(x) \pi_\nu(a | x)$ recovers ν , hence the expected feature and entropy objectives match those of $(\text{OPT})_2$.

Thus, $\pi_\nu \in \Pi$ is feasible for $(\text{OPT})_1$, and both problems yield the same objective value under this correspondence. $\square$

The above theorem establishes the equivalence of $(\text{OPT})_1$ and $(\text{OPT})_2$. In the remainder of this paper, our goal is to find the optimal solution to $(\text{OPT})_2$. To achieve this, we first derive

the dual of $(\text{OPT})_2$ and then demonstrate that the objective function in this dual formulation is both strongly convex and smooth. This property allows us to apply gradient descent with a constant learning rate to efficiently solve the problem.

VI. DUAL FORMULATION AND GRADIENT DESCENT ALGORITHM

In the remainder of this section, we focus on developing a gradient descent algorithm to find the optimal solution for $(\text{OPT})_2$, since it is a convex optimization problem. To do this, we first derive the dual of $(\text{OPT})_2$ and show that the objective function in the dual formulation is strongly convex and smooth. This property allows us to apply gradient descent with a constant learning rate, significantly reducing computational complexity.

Theorem 2. The dual of $(\text{OPT})_2$ is

$$\min_{\substack{\alpha \in \mathbb{R}^k \\ \beta, \theta \in \mathbb{R}^{\mathcal{X}}}} \left\{ \log \sum_{(x,a) \in \mathcal{X} \times \mathcal{A}} e^{k_{\alpha, \beta, \theta}(x,a)} - \langle \alpha, \langle f \rangle_{\pi_E, \mu_E} \rangle - \sum_{x \in \mathcal{X}} \theta_x \mu_E(x) \right\}. \quad (2)$$

Sketch of Proof. Starting from the min-max form in Remark 2, Sion's minimax theorem [16] lets us swap max and min. Applying the variational formula of [17, Prop. 1.4.2] then gives the dual objective in (2), so $(\text{OPT})_2$ has no duality gap.

For any fixed multipliers (α, β, θ) , the maximiser of

$$H(\nu) + \sum_{(x,a) \in \mathcal{X} \times \mathcal{A}} k_{\alpha, \beta, \theta}(x,a) \nu(x,a) \quad (3)$$

is the Boltzmann distribution $\nu^*(x,a) := \frac{e^{k_{\alpha, \beta, \theta}(x,a)}}{Z_{\alpha, \beta, \theta}}$, where $Z_{\alpha, \beta, \theta} := \sum_{(x,a) \in \mathcal{X} \times \mathcal{A}} e^{k_{\alpha, \beta, \theta}(x,a)}$, again by [17, Prop. 1.4.2]. $\square$

From now on, we define the objective function in (2), which represents the dual formulation of $(\text{OPT})_2$, as $h(\alpha, \beta, \theta)$. It is important to note that (3) is linear in (α, β, θ) ; and that the corresponding maximized value over ν ,

$$\log \sum_{(x,a) \in \mathcal{X} \times \mathcal{A}} e^{k_{\alpha, \beta, \theta}(x,a)},$$

is a convex function of (α, β, θ) . Since the remaining terms in h are linear in (α, β, θ) , we can conclude that h is a convex function.

Thus, the dual formulation of $(\text{OPT})_2$ is an unconstrained convex optimization problem in the variables (α, β, θ) , which can be solved using a gradient descent algorithm. Before we present the algorithm, we first demonstrate the smoothness and strong convexity of h . This ensures that the gradient descent algorithm will converge at a guaranteed rate when using a constant step size.

Theorem 3. h is L -smooth where $L := 2M^2 \sqrt{|\mathcal{X}| \cdot |\mathcal{A}|}$, here M can be determined explicitly.

Sketch of Proof. Compute the three partial gradients of h ; each is a sum over (x, a) weighted by the occupancy measure ν^* . Hence ∇h is Lipschitz as soon as ν^* is. Differentiate $\nu^*(x, a)$ with respect to any parameter $\gamma \in \{\alpha, \beta, \theta\}$ to obtain

$$\nabla_\gamma \nu^*(x, a) = \nu^*(x, a) (\nabla_\gamma k - \langle \nabla_\gamma k \rangle_{\nu^*}).$$

Uniform bounds on $\nabla_\gamma k$ give constants $M_\alpha, M_\beta, M_\theta$; let $M = \max\{M_\alpha, M_\beta, M_\theta\}$. Then $\|\nabla_\gamma \nu^*\| \leq 2M$, so ν^* is $2M$ -Lipschitz. Plugging this into the expressions for ∇h yields

$$\|\Delta(\nabla_\gamma h(\alpha, \beta, \theta))\| \leq 2M^2 \|\Delta(\alpha, \beta, \theta)\| \sqrt{|\mathcal{X}| \cdot |\mathcal{A}|},$$

where $\Delta(\cdot)$ denotes the difference between two evaluations at parameter points (α, β, θ) and $(\alpha', \beta', \theta')$. So h is L -smooth with $L = 2M^2 \sqrt{|\mathcal{X}| |\mathcal{A}|}$. $\square$

In the following theorem, we establish the strong convexity of h , under an additional assumption.

Theorem 4. *Suppose that the set*

$$\{(f(x, a, \mu_E), p(\cdot | x, a, \mu_E), e(\cdot | x, a)) : (x, a) \in \mathcal{X} \times \mathcal{A}\}$$

spans the space $\mathbb{R}^k \times \mathbb{R}^{\mathcal{X}} \times \mathbb{R}^{\mathcal{X}}$, where $e(y | x, a) := \mathbf{1}_{\{x=y\}}$. Then the function h is $\rho(D)$ -strongly convex over any compact subset $D \subset \mathbb{R}^m$, where $m = k + 2|\mathcal{X}|$.

Sketch of Proof. By computing the partial second derivatives, we observe that the Hessian of h with respect to (α, β, θ) coincides with the covariance matrix of the random vector

$$\mathbf{X}(x, a) := (f(x, a, \mu_E), p(\cdot | x, a, \mu_E), e(\cdot | x, a)), \quad (4)$$

under the distribution $(x, a) \sim \nu^*$. That is, $\text{Hes}(h)(\alpha, \beta, \theta) = \text{Cov}(\mathbf{X})$.

Assume, for contradiction, that $\text{Cov}(\mathbf{X})$ is not positive definite. Then there exists $\mathbf{v} \neq 0$ with $\mathbf{v}^\top \text{Cov}(\mathbf{X}) \mathbf{v} = \text{Var}(\langle \mathbf{v}, \mathbf{X} \rangle) = 0$, so $\langle \mathbf{v}, \mathbf{X} \rangle$ is almost surely constant and $\text{supp}(\text{Law}(\mathbf{X}))$ lies in the hyperplane $\{\mathbf{d} \in \mathbb{R}^m \mid \langle \mathbf{v}, \mathbf{d} \rangle = c\}$, a set of dimension at most $m - 1$. However, the map (4) spans $\mathbb{R}^m$ by assumption, and the Boltzmann policy ν^* assigns positive probability to every (x, a) . Hence the push-forward measure $\text{Law}(\mathbf{X})$ cannot be supported in a lower-dimensional subspace, a contradiction. Therefore, $\text{Cov}(\mathbf{X})$ must be positive definite.

The map $(\alpha, \beta, \theta) \mapsto \text{Cov}(\mathbf{X})$ is continuous. By the Extreme Value Theorem, on any compact $D \subset \mathbb{R}^m$ its smallest eigenvalue $\lambda_{\min}(\mathbf{X})$ attains a positive minimum: $\min_{(\alpha, \beta, \theta) \in D} \lambda_{\min}(\mathbf{X}) =: \lambda_{\min}(D) > 0$.

Using the bound above, $\nabla^2 h(\alpha, \beta, \theta) \succeq \rho(D) I_m$ for all $(\alpha, \beta, \theta) \in D$, where $\succeq$ stands for semidefinite order and $\rho(D) := \lambda_{\min}(D)$. Thus h is $\rho(D)$ -strongly convex on D . $\square$

Now we can introduce the gradient descent algorithm for finding the minimizer of h as follows.

Algorithm 1 Gradient Descent for Parameter Optimization

Require: Initial parameters $(\alpha_0, \beta_0, \theta_0)$, step size $\delta > 0$, number of iterations K

- 1: Set $(\alpha, \beta, \theta) \leftarrow (\alpha_0, \beta_0, \theta_0)$
- 2: **for** $k = 0, 1, \dots, K - 1$ **do**
- 3: $(\alpha, \beta, \theta) \leftarrow (\alpha, \beta, \theta) - \delta \nabla h(\alpha, \beta, \theta)$
- 4: **end for**
- 5: Compute $\nu_{\alpha, \beta, \theta}^*$ from the final parameters.
- 6: **return** (α, β, θ) and $\nu_{\alpha, \beta, \theta}^*$

Theorem 5. *Suppose that the step-size in gradient descent algorithm satisfies $0 < \delta \leq \frac{1}{L}$. Define the following compact subset of $\mathbb{R}^m$:*

$$D := \left\{ \mathbf{d} \in \mathbb{R}^m : \|\mathbf{d} - (\alpha_*, \beta_*, \theta_*)\| \leq \|(\alpha_0, \beta_0, \theta_0) - (\alpha_*, \beta_*, \theta_*)\| \right\}.$$

Then, for any k , we have

$$\begin{aligned} & \|(\alpha_k, \beta_k, \theta_k) - (\alpha_*, \beta_*, \theta_*)\|^2 \\ & \leq (1 - \delta \rho(D))^k \|(\alpha_0, \beta_0, \theta_0) - (\alpha_*, \beta_*, \theta_*)\|^2. \end{aligned}$$

Therefore, since $\nu_{\alpha, \beta, \theta}^(x, a)$ is $2M$ -Lipschitz continuous with respect to (α, β, θ) for all $(x, a) \in \mathcal{X} \times \mathcal{A}$, we also have*

$$\begin{aligned} & \|\nu_{\alpha_k, \beta_k, \theta_k}^* - \nu_{\alpha_*, \beta_*, \theta_*}^*\| \\ & \leq \sqrt{|\mathcal{X}| |\mathcal{A}|} 2M (1 - \delta \rho(D))^{k/2} \|(\alpha_0, \beta_0, \theta_0) - (\alpha_*, \beta_*, \theta_*)\|. \end{aligned}$$

Proof. This is a well-known result in convex optimization, so we omit the details. $\square$

We now relate the minimizer of h to the solution of the maximum entropy IRL problem. If $(\alpha_*, \beta_*, \theta_*)$ minimizes h (e.g., via gradient descent), then the optimal solution to **(OPT)₁** is:

$$\nu_{\alpha_*, \beta_*, \theta_*}^*(x, a) = \frac{e^{k \alpha_*, \beta_*, \theta_*(x, a)}}{\sum_{(x, a) \in \mathcal{X} \times \mathcal{A}} e^{k \alpha_*, \beta_*, \theta_*(x, a)}}.$$

The policy corresponding to this distribution is:

$$\pi_{\nu_{\alpha_*, \beta_*, \theta_*}^*}(a | x) = \frac{\nu_{\alpha_*, \beta_*, \theta_*}^*(x, a)}{\nu_{\alpha_*, \beta_*, \theta_*}^*(x)}.$$

This policy solves the maximum causal entropy problem **(OPT)₁**.

VII. NUMERICAL ILLUSTRATION

We illustrate our methodology using a malware spread model [18] under an average-cost¹ criterion. Agents possess local states $x_t \in \{0, 1\}$, representing healthy (0) or infected (1) conditions. At each step, agents select actions $a_t \in \{0, 1\}$, where 0 corresponds to “do nothing” and 1 to “repair”. The system dynamics follow

$$x_{t+1} = \begin{cases} x_t + (1 - x_t)w_t, & \text{if } a_t = 0, \\ 0, & \text{if } a_t = 1, \end{cases}$$

where $w_t \sim \text{Bernoulli}(q = 0.9)$. Thus, a healthy agent that does nothing becomes infected with probability q , whereas repair guarantees recovery.

¹Here, we formulate the problem as a cost minimization rather than a reward maximization, but the two are equivalent.

The per-stage cost is given by $c(x, a, \mu) = \theta_1 x + \theta_2 x \mu(1) + \theta_3 a$, with $\theta = [-2, -1, -3]$. Here, $\mu(1)$ denotes the fraction of infected agents, θ_3 the repair cost, and $\theta_1 + \theta_2 \mu(1)$ the risk associated with infection.

We first address the forward problem by computing the MFE (π_E, μ_E) , obtained numerically as

$$\pi_E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mu_E = \begin{bmatrix} 0.5263 \\ 0.4737 \end{bmatrix}.$$

This equilibrium indicates that healthy agents do nothing, whereas infected agents always choose repair.

Using this equilibrium (π_E, μ_E) , we generate the average feature expectation vector (1) where $f(x, a, \mu(1)) = (x, x\mu(1), a)$. Our goal is to recover (π_E, μ_E) using IRL.

We solve the dual IRL problem to recover the optimal joint distribution $\nu(x, a)$, from which the policy was obtained via $\pi(a|x) = \frac{\nu(x, a)}{\mu_E(x)}$. The resulting policy is

$$\pi = \begin{bmatrix} 0.9709 & 0.0291 \\ 0.0303 & 0.9697 \end{bmatrix} \quad \text{versus} \quad \pi_E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The recovered strategy closely approximated the expert behavior. The convergence of the dual objective h is illustrated in Fig. 1, confirming the numerical consistency of the algorithm.

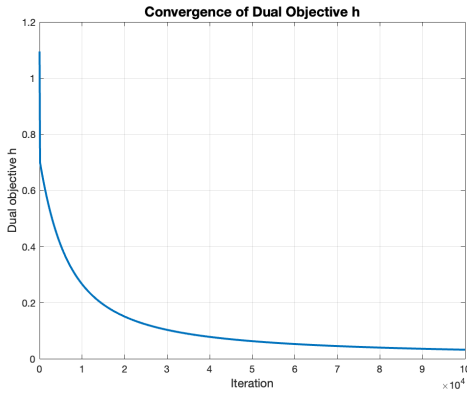


Fig. 1. Convergence of dual objective h over iterations.

VIII. CONCLUSION

In conclusion, we have explored the inverse reinforcement learning (IRL) problem in discrete-time, infinite-horizon mean-field games (MFGs) with an average-reward criterion. Unlike the forward problem, IRL relies on expert demonstrations that are optimal under an unknown reward. Our objective is to recover a policy that explains the expert behavior under some unknown reward. To this end, we proposed a convex reformulation based on occupation measures and analyzed its dual, showing that the resulting objective is smooth and strongly convex. This structure enabled the use of a simple first-order algorithm to recover an entropy-maximizing policy consistent with expert behavior. The numerical results support the stability and effectiveness of the approach. Overall, our formulation provides a tractable basis for IRL in large-population systems and may serve as a starting point for extending the framework to more general or complex settings.

REFERENCES

- [1] M. Huang, R. Malhamé, and P. Caines, “Large population stochastic dynamic games: Closed loop McKean-Vlasov systems and the Nash certainty equivalence principle,” *Communications in Information Systems*, vol. 6, pp. 221–252, 2006.
- [2] J. Lasry and P. Lions, “Mean field games,” *Japan. J. Math.*, vol. 2, pp. 229–260, 2007.
- [3] G. Weintraub, L. Benkard, and B. Van Roy, “Oblivious equilibrium: A mean field approximation for large-scale dynamic games,” in *Advances in Neural Information Processing System*, Jan. 2005.
- [4] S. Adlakha, R. Johari, and G. Weintraub, “Equilibria of dynamic games with many players: Existence, approximation, and market structure,” *Journal of Economic Theory*, vol. 156, pp. 269–316, 2015.
- [5] M. Lauriere, S. Perrin, J. Perolat, S. Girgin, P. Muller, R. Elie, M. Geist, and O. Pietquin, “Learning in mean field games: A survey,” 2024.
- [6] S. Adams, T. Cody, and P. A. Beling, “A survey of inverse reinforcement learning,” *Artif. Intell. Rev.*, vol. 55, no. 6, pp. 4307–4346, Aug. 2022.
- [7] J. Yang, X. Ye, R. Trivedi, X. Hu, and H. Zha, “Learning deep mean field games for modelling large population behaviour,” arXiv:1711.03156, 2018.
- [8] Y. Chen, L. Zhang, J. Liu, and S. Hu, “Individual-level inverse reinforcement learning for mean field games,” in *Proceedings of the 21st International Conference on Autonomous Agents and Multiagent Systems*, ser. AAMAS ’22, 2022, pp. 253–262.
- [9] Y. Chen, L. Zhang, J. Liu, and M. Witbrock, “Adversarial inverse reinforcement learning for mean field games,” ser. AAMAS ’23, 2023, pp. 1088–1096.
- [10] G. Ramponi, P. Kolev, O. Pietquin, N. He, M. Lauriere, and M. Geist, “On imitation in mean-field games,” 2023.
- [11] Z. Zhou, M. Bloem, and N. Bambos, “Infinite time horizon maximum causal entropy inverse reinforcement learning,” *IEEE Transactions on Automatic Control*, vol. 63, no. 9, pp. 2787–2802, 2018.
- [12] B. Anahtarci, C. D. Kariksiz, and N. Saldi, “Maximum causal entropy inverse reinforcement learning for mean-field games,” 2024. [Online]. Available: <https://arxiv.org/abs/2401.06566>
- [13] S. P. Meyn and R. L. Tweedie, *Markov Chains and Stochastic Stability*, 2nd ed. Cambridge, UK: Cambridge University Press, 2009.
- [14] B. D. Ziebart, A. L. Maas, J. A. Bagnell, and A. K. Dey, “Maximum entropy inverse reinforcement learning,” in *Proceedings of the 23rd AAAI Conference on Artificial Intelligence*. AAAI Press, 2008, pp. 1433–1438.
- [15] F. Alajaji and P.-N. Chen, *An Introduction to Single-User Information Theory*. Cham, Switzerland: Springer, 2018.
- [16] M. Sion, “On general minimax theorems,” *Pacific Journal of Mathematics*, vol. 8, pp. 171–176, 1958.
- [17] P. Dupuis and R. Ellis, *A Weak Convergence Approach*

to the Theory of Large Deviations. Wiley-Interscience, 1997.

- [18] J. Subramanian and A. Mahajan, “Reinforcement learning in stationary mean-field games.” International Foundation for Autonomous Agents and Multiagent Systems, 2019, pp. 251–259.